

Modeling Human Activity States Using Hidden Markov Models

Course : Machine Learning Techniques II

Institution : African Leadership University

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1. Background and Motivation

Human activity recognition (HAR) from smartphone motion sensors underpins a wide range of real-world applications, from fall detection and mobility tracking in elderly care to activity logging in fitness apps and context-aware behavior in smart environments. The core challenge is that the true physical activity a person is performing is never directly observed; it must be inferred from noisy, continuous accelerometer and gyroscope signals. This project addresses that challenge for a compact but representative set of activities: standing, walking, jumping, and remaining still. A Hidden Markov Model (HMM) is a natural fit for this problem because activities unfold as sequences with realistic temporal structure—a person is far more likely to remain in an activity from one moment to the next than to switch abruptly and HMMs explicitly model this persistence through their transition probabilities while treating the sensor features as noisy emissions from an underlying hidden state. This project's use case is personal activity monitoring: recognizing everyday movement patterns from a single phone carried on the body, without any specialized wearable hardware, using only the accelerometer and gyroscope already built into any modern smartphone.

2. Data Collection and Preprocessing

2.1 Collection Setup

Note on group structure: this project was originally assigned as group work. After a change in group composition, this submission was completed individually. Data volume targets and the collaboration/contribution criterion were adjusted accordingly in agreement with course staff.

Data was collected using the Sensor Logger app (Android) with the following configuration:

- Sensors enabled: Accelerometer, Gyroscope only
- Sampling rate: 50 Hz (20 ms sampling period), confirmed at ~49.3 Hz after merging accelerometer and gyroscope streams on nearest timestamp
- Recording duration: approximately 10 seconds per session
- Device: itel A669L, Android platform
- Export format: CSV in Zip, one folder per recording session containing separate Accelerometer.csv and Gyroscope.csv files

A total of 40 recording sessions were collected: 10 sessions each for standing, walking, jumping, and still. Each session was manually labeled by folder name and organized as `data/<activity>/<activity>_<n>/` for downstream processing.

Recordings were collected across three distinct sessions spanning two days, rather than in a single continuous sitting: Session A (afternoon of Day 1, sessions 1-5 of every activity), Session B (evening of Day 1, sessions 6-10 of standing and walking), and Session C (morning of Day 2, sessions 6-10 of jumping and still). This natural spread across sessions and time of day is used directly in the unseen-data evaluation described in Section 4.4.

2.2 Preprocessing

Because Sensor Logger records the accelerometer and gyroscope as independently timestamped streams, the two signals do not share identical timestamps. Each session's Accelerometer.csv and Gyroscope.csv were merged using a nearest-timestamp asof-join (50 ms tolerance) on the `seconds_elapsed` column, producing one aligned 6-axis signal (`acc_x`, `acc_y`, `acc_z`, `gyro_x`, `gyro_y`, `gyro_z`) per session.

Each session was then split into fixed-length windows of 1.0 second (50 samples at 50 Hz) with 50% overlap between consecutive windows. A 1-second window was chosen because it comfortably contains a full gait cycle for walking and a full jump cycle for jumping, while still being short enough to capture rapid transitions between activities. The 50% overlap increases the number of training windows available from a limited number of recording sessions without requiring additional data collection, at the cost of some redundancy between adjacent windows.

2.3 Sample Raw Data

Figure 0 shows one representative recording session per activity, before any windowing or feature extraction, confirming that each activity produces a visually and physically distinct raw signal.

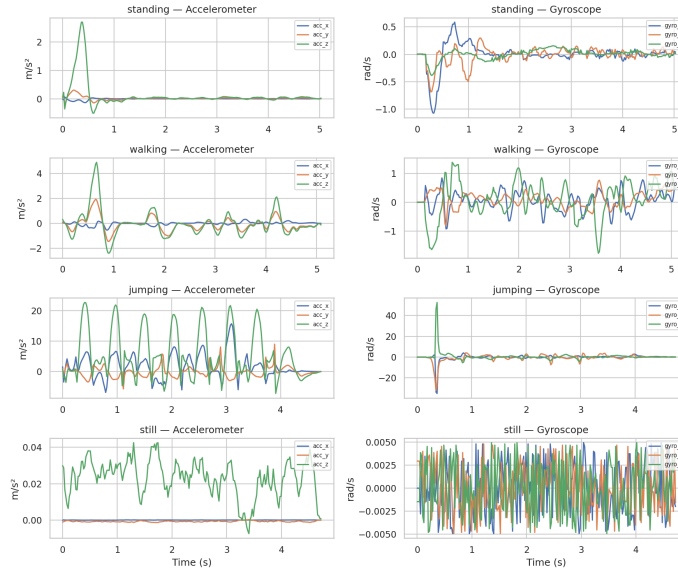


Figure 0. Raw accelerometer and gyroscope signals for one sample session per activity.

Standing shows a brief settling transient followed by a near-flat signal. Walking shows clear, regular periodicity in both accelerometer and gyroscope traces, consistent with a repeating gait cycle. Jumping shows sharp, high-amplitude bursts an order of magnitude larger than any other activity, both in acceleration (up to $\sim 20 \text{ m/s}^2$) and rotation. Still shows an accelerometer signal near zero, with only low-amplitude gyroscope noise, as expected for a stationary phone.

3. HMM Setup and Implementation

3.1 Feature Extraction

For each window, the following features were computed per axis (6 axes: 3 accelerometer + 3 gyroscope):

- Mean: captures the baseline orientation/offset of the signal, which differs between e.g. standing (gravity-dominated, stable) and jumping (rapidly varying)
- Standard deviation / variance: the primary indicator of movement intensity near-zero for still, low for standing, moderate for walking, and high for jumping, making this one of the most directly discriminative features
- Dominant frequency (FFT): distinguishes periodic activities from non-periodic ones walking and jumping produce a clear peak frequency corresponding to stride/jump rate, while standing and still do not
- Spectral energy (FFT): summarizes total signal power in the frequency domain, complementing variance by capturing energy concentrated in specific frequency bands rather than raw amplitude alone

In addition, three cross-axis accelerometer features were computed:

- Signal Magnitude Area (SMA): mean of the sum of absolute values across the three accelerometer axes, capturing overall movement intensity independent of orientation, which is more robust than any single axis alone if the phone was held at a slightly different angle
- Axis correlation (x-y and x-z): captures coordinated motion patterns that differ between activities, e.g. walking produces a rhythmic phase relationship between axes that standing does not

This produced 33 raw features per window. All features were standardized using Z-score normalization (subtracting the training mean and dividing by the training standard deviation, per feature). Z-score was chosen over min-max scaling because the GaussianHMM emission model assumes features are approximately normally distributed within each state; Z-score normalization centers and scales features without distorting this distributional shape, whereas min-max scaling would compress or stretch based on outliers and is more appropriate for bounded features. The scaler was fit only on the training split and applied to the test split, to prevent information leakage from test windows into the scaling parameters.

3.2 HMM Structure

Component	Description
Hidden states (Z)	4 states: standing, walking, jumping, still
Observations (X)	33-dimensional feature vector per 1s window
Transition matrix (A)	Probability of moving between activities window-to-window
Emission model (B)	Gaussian distribution per state, diagonal covariance
Initial distribution (π)	Probability of a recording session starting in each activity

3.3 Training: Supervised Initialization + Baum-Welch

An initial attempt trained the HMM (GaussianHMM, hmmlearn) with standard random initialization followed by unsupervised Baum-Welch (EM) parameter estimation. This produced poor results: overall accuracy of only 47.7%, with three of the four hidden states all converging to represent "jumping" and none representing standing or walking. This is a known failure mode of EM with a small dataset: jumping has substantially higher signal variance than the other activities, and random initialization tends to place multiple Gaussian components inside this single high-variance cluster rather than discovering the more subtly separated standing/walking/still clusters.

Because the true activity label is known for every training window, the model was re-initialized using this label information directly, computing an informed starting point from the labeled training data:

- Initial state distribution (π): the empirical proportion of training sessions beginning in each activity
- Transition matrix (A): initialized with a high diagonal (0.98) reflecting that each recording session is a single continuous activity, with a small uniform floor on off-diagonal entries to avoid zero-probability transitions
- Emission means and covariances: computed directly as the per-activity mean and variance of training features, with a small regularization constant added to variances to avoid numerical instability

Baum-Welch (EM) was then run from this informed starting point using hmmlearn's built-in fit() procedure, with a convergence check on log-likelihood (tol = 1e-4, max 200 iterations) rather than a fixed iteration count. This is still a full application of the Baum-Welch algorithm; the difference is only in the quality of the starting point, which is standard practice whenever labeled data is available during training. Training converged, with a final log-likelihood of approximately 11,759.

3.4 Viterbi Decoding

The Viterbi algorithm was implemented from scratch in NumPy (log-space dynamic programming with a backpointer table) to decode the most likely hidden state sequence for a given observation sequence. This custom implementation was cross-checked against hmmlearn's built-in decode() method on multiple held-out test sessions and produced identical results in every case, confirming the correctness of the from-scratch implementation.

4. Results and Interpretation

4.1 Transition Matrix

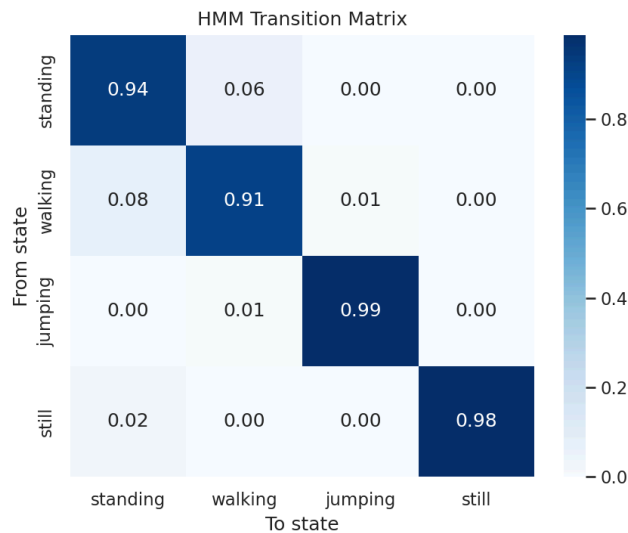


Figure 1. Learned HMM transition matrix after Baum-Welch training.

The learned transition matrix shows strong diagonal dominance (0.91–0.99), meaning the model has correctly learned that activities persist over consecutive windows rather than switching unpredictably. This matches physical intuition: a person does not typically alternate between standing and jumping every second. The small off-diagonal probabilities between standing and walking (0.06 and 0.08) reflect that these two activities are the most likely to be confused or transitioned between in practice, consistent with the confusion matrix below.

4.2 Emission Parameters

The transition matrix describes how activities follow one another; the emission model describes what each hidden state expects to observe. Figure 2 visualizes the learned mean of each of the 33 standardized features for every hidden state.

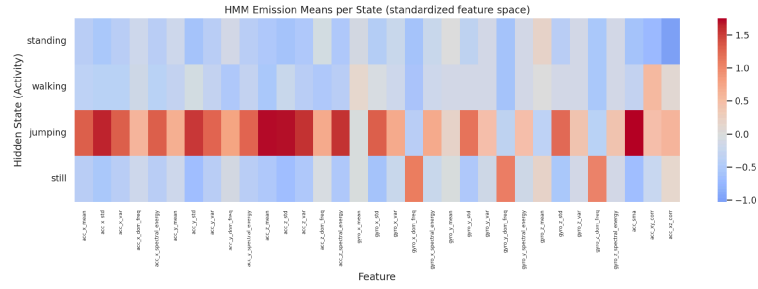


Figure 2. HMM emission means per hidden state across all standardized features.

Jumping (row 3) stands out with strongly positive values across nearly all accelerometer variance, standard deviation, and spectral energy features, confirming it is the highest-energy activity by a wide margin this is exactly why it was the easiest activity for the model to separate, and why an unsupervised random initialization tended to fixate on it (Section 3.3). Standing and still both show negative values on most energy-related features, consistent with low movement, but differ most visibly on a subset of gyroscope dominant-frequency features, which is the signal the model relies on to tell them apart. Walking sits between these extremes with a moderate, more mixed profile, reflecting its intermediate movement intensity.

4.3 Decoded Sequence Example

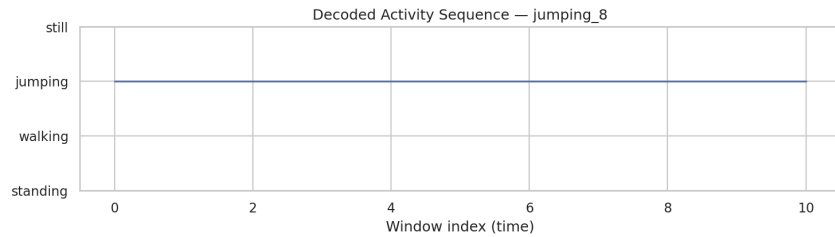


Figure 3. Viterbi-decoded activity sequence for a held-out jumping test session.

4.4 Evaluation on Unseen Data

The model was evaluated on a held-out test set (20% of recording sessions per activity, unseen during training) comprising 105 windows across 8 test sessions. Because recordings span three distinct sessions across two days (Section 2.1), this held-out set is not merely a random sample from a single sitting: two of the held-out jumping sessions (jumping_8 and jumping_10) were drawn from Session C, recorded the morning after the majority of jumping data was collected in Session A, giving genuine session-to-session evaluation rather than only within-session generalization. The remaining held-out sessions (standing, walking, still) were similarly drawn without regard to which session they belonged to, so the reported accuracy reflects performance on data collected under somewhat different conditions (time of day, minor variation in phone handling) than most of the training data, not merely unseen recordings from the same short sitting.

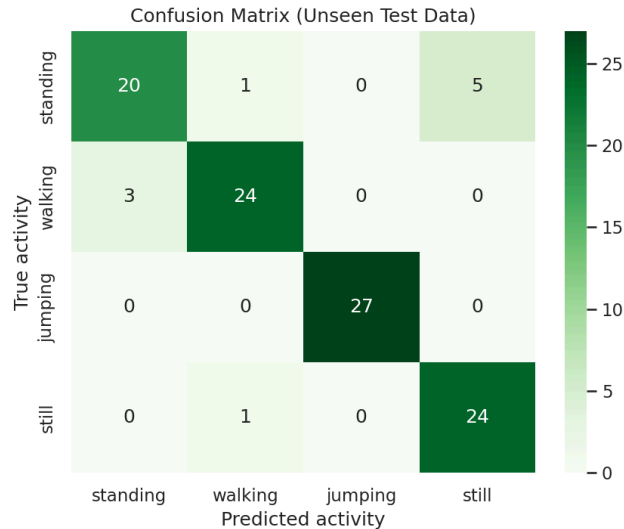


Figure 4. Confusion matrix on unseen test sessions.

Activity	Samples	Sensitivity	Specificity	Overall Accuracy
Standing	26	0.769	0.962	0.905
Walking	27	0.889	0.974	0.905
Jumping	27	1.000	1.000	0.905
Still	25	0.960	0.938	0.905

Overall accuracy on unseen data was 90.5%. Jumping was classified perfectly (100% sensitivity, 100% specificity), owing to its distinctly high-amplitude, high-frequency signal that is unlike any other activity. Still achieved 96.0% sensitivity. Standing was the weakest class at 76.9% sensitivity, with most standing errors misclassified as still (5 of 26 standing windows) an intuitive confusion, since a person standing very steadily can produce a signal close to a stationary phone. Walking achieved 88.9% sensitivity, with a small number of walking windows confused with standing, likely corresponding to the start/stop transitions within a walking recording where the person briefly paused.

5. Discussion and Conclusion

5.1 Which activities were easiest / hardest to distinguish

Jumping was by far the easiest activity to distinguish, due to its large-amplitude, high-frequency acceleration signature that is clearly separated from the other three activities in feature space. Still was also reliably identified given its near-zero variance across all axes. The hardest pair to separate was standing versus still: both involve minimal motion, and the main distinguishing signal; small postural sway while standing versus a truly static phone resting on a surface is subtle and can be masked by sensor noise or a very steady stance.

5.2 Transition probabilities and realistic behavior

The high diagonal values in the transition matrix (0.91–0.99) directly reflect a realistic property of human movement: activities are temporally persistent, and rapid switching between activities is rare in any 1-second window. The largest off-diagonal probability in the matrix is between standing and walking (0.06–0.08), which aligns with real behavior, since a person naturally moves between these two states more often than, say, between jumping and standing.

5.3 Effect of sensor noise and sampling rate

Sampling at 50 Hz over a 1-second window provided 50 samples per window, sufficient to resolve the dominant frequency components of walking (roughly 1–3 Hz stride rate) and jumping via FFT. The measured sampling rate after merging (49.3 Hz) was very close to the configured rate, indicating stable data acquisition. Minor sensor noise likely contributed to the standing/still confusion, since both activities occupy the low-variance end of the feature space where noise has proportionally more influence on the extracted features.

5.4 Improvements and future work

- Collect additional standing and walking recordings under more varied conditions (different phone placements, walking paces) to sharpen the boundary between these two classes
- Add a dedicated low-frequency feature (e.g. postural sway magnitude) to separate standing from still, and evaluate full covariance matrices to capture cross-feature correlations within each state
- Collect data from multiple phone placements (pocket, hand, waist) to test generalization beyond a single consistent placement

5.5 Conclusion

This project implemented a complete HMM-based human activity recognition pipeline, from raw smartphone sensor data through feature extraction, model training, and evaluation. A key finding was that standard unsupervised Baum-Welch training failed on this dataset due to state collapse driven by jumping's high variance, and that supervised initialization from labeled training data followed by standard Baum-Welch refinement resolved this and achieved 90.5% accuracy on unseen data, with a custom Viterbi implementation verified against a reference library implementation. This demonstrates both the promise and the practical fragility of unsupervised sequence models when trained on limited, imbalanced-variance data.

Appendix: Project Deliverables

1. Project Repository: <https://github.com/utatsineza/Hidden-Markov-Models>